

**COLUMN BUCKLING OF COMPOSITE
BEAMS**

Section: **3.4 – STRENGTH OF COMPOSITE
PARTS**

Chapter: **3440**

Title: **COLUMN BUCKLING OF COMPOSITE
BEAMS**

Revision: **1.1**

Date: **08/06/2020**

1 INTRODUCTION

1.1 OBJECT

This chapter presents the methodology to be used for calculation of the strength of beams loaded in compression.

1.2 SCOPE AND LIMITATIONS

Column buckling of straight beams of uniform cross section made of fibrous composite laminates.

Out of the scope of this chapter:

- ✓ Column buckling of inelastic isotropic materials (see chapter 3140)
- ✓ Instabilities associated to torsion (at least in this revision)
- ✓ Beam-column interaction (at least in this revision)

2 METHOD DESCRIPTION

A column may fail in one of the following modes and therefore the strength of a column is the lowest strength associated with any of the following failure modes:

- ✓ Flexural instability
- ✓ Crippling
- ✓ Interaction between local crippling and flexural instability
- ✓ Torsional instability. This is relatively rare in columns but is to be expected, in general in open thin walled sections having wide flanges and short column lengths

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DEFENCE AND SPACE

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Previous wording is mostly an extract of Niu's book, chapter 10, applicable specifically for beams made of isotropic materials. For beams made of fibrous composite laminates, typically, at least, the compression failure may be also caused by:

- ✓ Residual strength after barely visual impact damage

Only flexural instability and its interaction with crippling shall be treated in this chapter. Crippling analysis is treated in chapter 3430, and strength after BVID in chapter 3405.

2.1 EULER EQUATION FOR BUCKLING OF (LONG) COLUMNS

The critical buckling load of a (long) column is given by the Euler formula, which can be written as follows:

$$P_E = C \cdot \pi^2 \cdot EI / L^2 \quad \text{or, as stress} \quad F_E = C \cdot \pi^2 \cdot EI / (A \cdot L^2)$$

...where:

- C is the end fixity coefficient
- EI is the bending stiffness of the cross section
- L is the length of the column

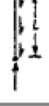
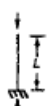
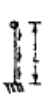
Column shape and end fixity		End fixity coefficient	Column shape and end fixity		End fixity coefficient
	Uniform column, axially loaded, pinned ends	$c = 1$ $\frac{1}{\sqrt{c}} = 1$		Uniform column, distributed axis load, one end fixed, one end free	$c = 0.794$ $\frac{1}{\sqrt{c}} = 1.12$
	Uniform column, axially loaded, fixed ends	$c = 4$ $\frac{1}{\sqrt{c}} = 0.5$		Uniform column, distributed axis load, pinned ends	$c = 1.87$ $\frac{1}{\sqrt{c}} = 0.732$
	Uniform column, axially loaded, one end fixed, one pinned end	$c = 2.05$ $\frac{1}{\sqrt{c}} = 0.7$		Uniform column, distributed axis load, fixed ends	$c = 7.5$ (Approx.) $\frac{1}{\sqrt{c}} = 0.365$
	Uniform column, axially loaded, one end fixed, one end free	$c = 0.25$ $\frac{1}{\sqrt{c}} = 2$		Uniform column, distributed axis load, one end fixed, one end pinned	$c = 6.08$ $\frac{1}{\sqrt{c}} = 0.406$

Figure 1. Values of end fixity coefficient for several support conditions, from Niu chapter 10.2

Calculation of the bending stiffness of composite beams is discussed in chapter 3410.

The same formula can be found in literature as function of the following magnitudes:

- Effective column length $L' = L / C^{0.5}$
- Radius of gyration of the section $\rho = (I/A)^{0.5} = (EI/EA)^{0.5}$

...where A is the area of the cross section and last format of the radius of gyration more appropriate for composite beams.

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2.2 INTERACTION BETWEEN COLUMN BUCKLING, CRIPPLING & CAI

In the short to intermediate column range, characterized by low slenderness ratios, failure is due to either local crippling or block compression of the material. Last failure mode, translated to composite beams, could be either plain compression strength or residual compression strength after impact in case the part must be damage tolerant.

Thus the cut-off value for the strength in the short column range shall be the minimum between:

- ✓ Crippling strength of the cross section
- ✓ For some section geometries including flanges with free edges at section extreme points, the local buckling load of that flange with a free edge (see figure below)

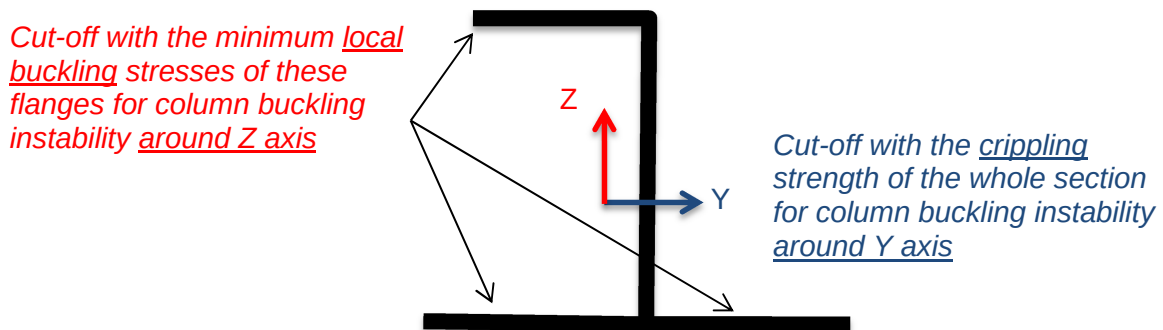


Figure 2. Crippling and flange local buckling cut-off

In other words, in case the section includes flange-segments with one of its sides free, if the flange is normal to the axis of minimum bending inertia, the cut-off value shall not be higher than the local buckling load of that flange. A typical example of this case is a tee profile bonded to a skin panel. It cannot be expected that the post-buckled web segment of a tee (a flange with one side free normal to the skin) provides bending inertia to the skin+stringer section.

- ✓ For parts with damage tolerant requirements, the minimum compression strength after BVID of any of the profile segments

To facilitate checking a column for failure in the flexural instability mode various empirical formulas have been developed from tests prediction short to intermediate column failure. The Johnson curve is recommended for the short to intermediate column lengths. Typically found in literature as:

$$F_c = F_{cr} - [F_{cr}^2 \cdot (L'/\rho)^2] / (4 \cdot \pi^2 \cdot E)$$

Expressed in a more convenient manner to deal with composite beams and alternative cut-off strengths not necessarily due to crippling:

$$F_c = \sigma_{co} - \sigma_{co}^2 / [(4 \cdot \pi^2 \cdot E) / (L'/\rho)^2] = \sigma_{co} - \sigma_{co}^2 / [(4 \cdot C \cdot \pi^2 \cdot EI) / (A \cdot L^2)] =$$

$$F_c = \sigma_{co} - \sigma_{co}^2 \cdot A \cdot L^2 / (4 \cdot C \cdot \pi^2 \cdot EI)$$

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This, the Johnson parabola shall be used within the range of zero length up to the length giving a compression allowable stress of half the cut-off stress. For greater lengths, the Euler formula shall be used.

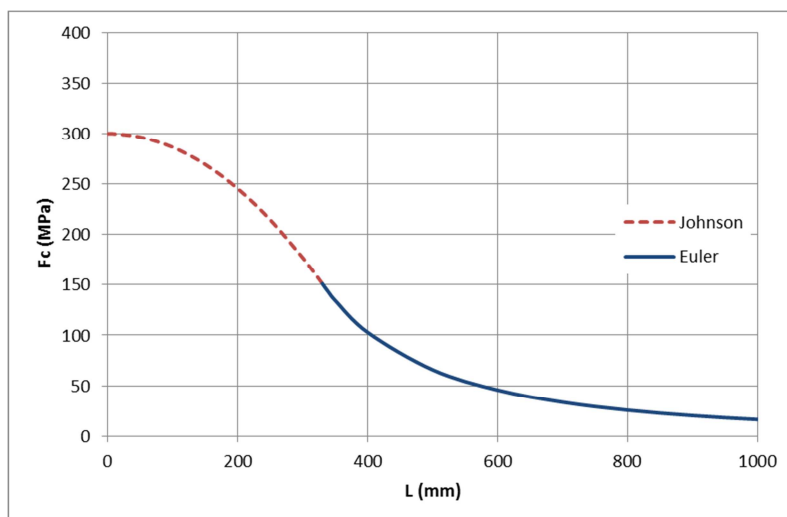


Figure 3. Example of Johnson-Euler fitting curve with its intersection at $\sigma_{co}/2$

Warning! Torsional instability not covered by this current revision (see discussion in first page).

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3 APPENDICES

ABBREVIATIONS AND SYMBOLS

<i>Symbol</i>	<i>Units</i>	<i>Definition</i>
A	mm ²	Area of the cross section
[A]	N/mm	In-plane stiffness matrix of a composite laminate
[B]	N	Matrix defining the laminate shear-bending coupling
BVID	-	Barely visual impact damage
C	-	Fixity coefficient
CAI, σ_{CAI}	MPa	Compression residual strength after BVID
[D]	Nmm	Bending stiffness matrix of a composite laminate
E	MPa	Young modulus
ϵ_x	-	Strain in longitudinal X direction (parallel to stringer axis)
F_c	MPa	Compression ultimate stress of the column
F_{cr}	MPa	Crippling stress of the stringer cross section
F_{st}	MPa	Stress of the stringer cross section
F_E	MPa	Buckling compression stress of the long column (Euler formula)
I	mm ⁴	Bending moment of inertia of the cross section
L	mm	Length of the composite beam
P_{cr}	N	Crippling allowable force of the cross section
P_E	N	Buckling compression force of the long column (Euler formula)
P_x	N	Longitudinal compression load applied on a stringer plus adjacent skin assembly
ρ	mm	Radius of gyration of the section, equal to $(I/A)^{0.5}$
σ_{co}	MPa	Cut-off stress for short beams in the Johnson parabola

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REFERENCES

	<i>Document Reference</i>	<i>Issue</i>	<i>Title</i>
Ref. 1	[E. F. Bruhn]	1973	ANALYSIS AND DESIGN OF FLIGHT VEHICLE STRUCTURES
Ref. 2	[M. C. Y. Niu]	2 nd Ed.	AIRFRAME STRESS ANALYSIS AND SIZING
Ref. 3	[Robert M. Jones]	2 nd Ed.	MECHANICS OF COMPOSITE MATERIALS

SOFTWARE

<i>Program</i>	<i>Version</i>	<i>Description</i>

Table 1. Software programs referenced in this chapter

RECORD OF REVISIONS

<i>Revision Date</i>	<i>Authors</i>	<i>Change Reason</i>	<i>Pages/Sub-chapters affected</i>
1.0 30/04/2020	G. Baños	First issue	All pages
1.1 08/06/2020	G. Baños	Removal of §2.3 "COLUMN BUCKLING OF STIFFENED PANELS", a topic left for the assembly-method chapters	Pages 4-5

Table 2. Record of Revisions: authors, change reasons and sections/pages affected

APPROVAL SIGNATURES TABLE

<i>Dpt. \ Site</i>	<i>Getafe</i>	<i>Manching</i>
Stress Methods (TEASP-TL3)	G. Baños 08/06/2020	Florian Glock 08/06/2020

Table 3. Approval signatures table for last revision of this chapter